

BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, PILANI
FIRST SEMESTER 2012-13
ME/MF F212: *Fluid Mechanics*

Time: 1.5hrs

Marks:25

Mid-Semester Examination (04/10/2012)

Instructions

1. Ethical practice is the norm. Defaulters will be severely penalised.
2. Open book: Only notes in your handwriting are allowed.
3. Illegible answers will not be assessed. Please box important milestones in the answers.
4. State all assumptions made in your solution.
5. Read the questions carefully and answer accordingly and succinctly. Skew, irrelevant answers will not be assessed but instead be penalised.
6. **IMPORTANT** Write your answers such that the rationale behind your thoughts is clearly presented. A major portion of credit (if not all) is reserved for this. Repeat, correct end answers do not assure you of any credit, rather the rationale that is articulately presented will take you places.

1 Fluid Statics

An opened heavy cap in the shape of a circular blunt nosed cone, with the angle α at the base and with the radius of the base R , stands on a horizontal plane (see fig 1). Of what weight G must the cap be to keep water poured into it up to the height H ? The pressure everywhere outside the cap equals the pressure in the atmosphere. [3]

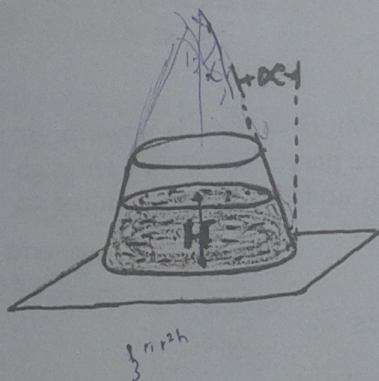


Figure 1: Question 1

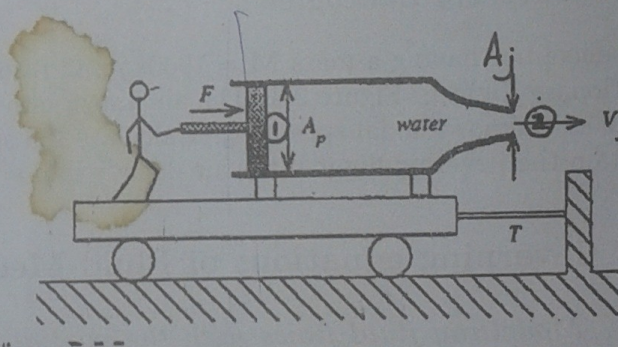


Figure 2: Question 2.1

2 Control Volume Analysis of Conservation Laws of Fluid Mechanics

2.1 Cart with Human Powered Rocket

A cylinder equipped with a frictionless piston of area A_p and filled with water of density ρ is mounted on a wheeled cart, as shown in Figure 2.1. A person standing on the cart exerts a force F on the piston, causing a water jet to be ejected into the surrounding air through the nozzle on the right of the cylinder. The cart is restrained from moving to the left by a rope tied to a pillar. Derive expressions for the velocity V_j of the water jet and the tension T in the rope in terms of ρ , F , A_j and A_p . Explain clearly why the orifice formula *can not* be used here.

Hint: Unlike the problem discussed in class, a good approximation is to use the Bernoulli equation between points (1) and (2) in the figure. [4]

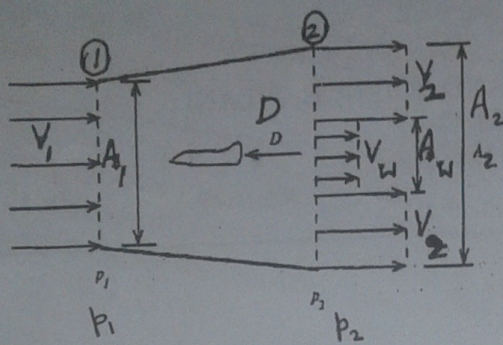


Figure 3: Question 2.2

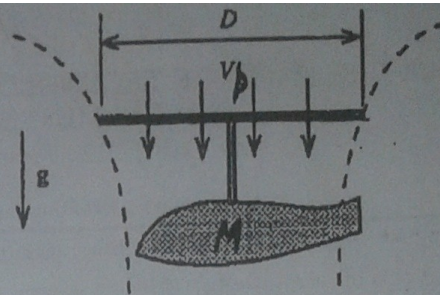


Figure 4: Question 2.3

2.2 Aeroplane drag via a Wind Tunnel Experiment

A variable-area wind tunnel is used to determine the drag force D on a model of an airplane. The upstream flow in the tunnel has a uniform speed V_1 and pressure p_1 all across the inlet area A_1 of the tunnel test section. Although the downstream pressure p_2 is also uniform across the outlet flow area A_2 , there is a core of the flow, of area A_w , within which speed V_w is less than the uniform speed V_2 in the remainder of the exiting flow.

The test engineer adjusts the exit area A_2 until the pressure p_2 and velocity V_2 match the inlet values, p_1 and V_1 . This will simulate free flight in the atmosphere because the streamlines passing far from the aircraft model are little disturbed. Assuming incompressible flow and negligible wall friction in the tunnel, derive expressions for the downstream area A_2 and the model drag force D in terms of A_1 , V_1 , A_w , V_w , p_1 and ρ , i.e., the quantities measured in the test by the test engineer. [4]

2.3 Hovering Helicopter

A helicopter, having a mass $M = 1000 \text{ kg}$ and rotor diameter $D = 10 \text{ m}$, is hovering in stationary flight in Figure 2.3. The air drawn into the rotor passes downward through it at an unknown axial speed V_p . Calculate the speed V_p and the power required to sustain the hovering flight if the air density is $\rho = 1.2 \text{ kg/m}^3$. [2]

3 Governing equations of Fluid Mechanics

3.1 Modelling fluid flows with the partial differential equations of Fluid Mechanics

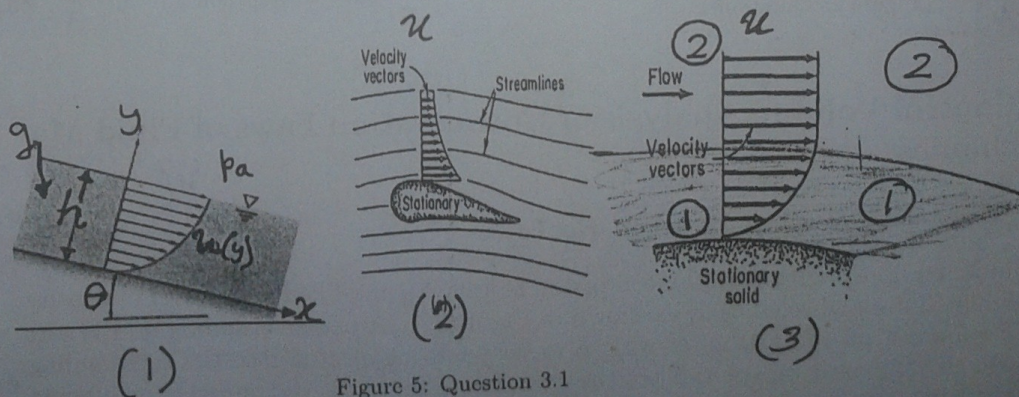


Figure 5: Question 3.1

Consider three important fluid flow problems. In each case, the velocity profiles calculated in steady state are shown in the figure. Study them carefully.

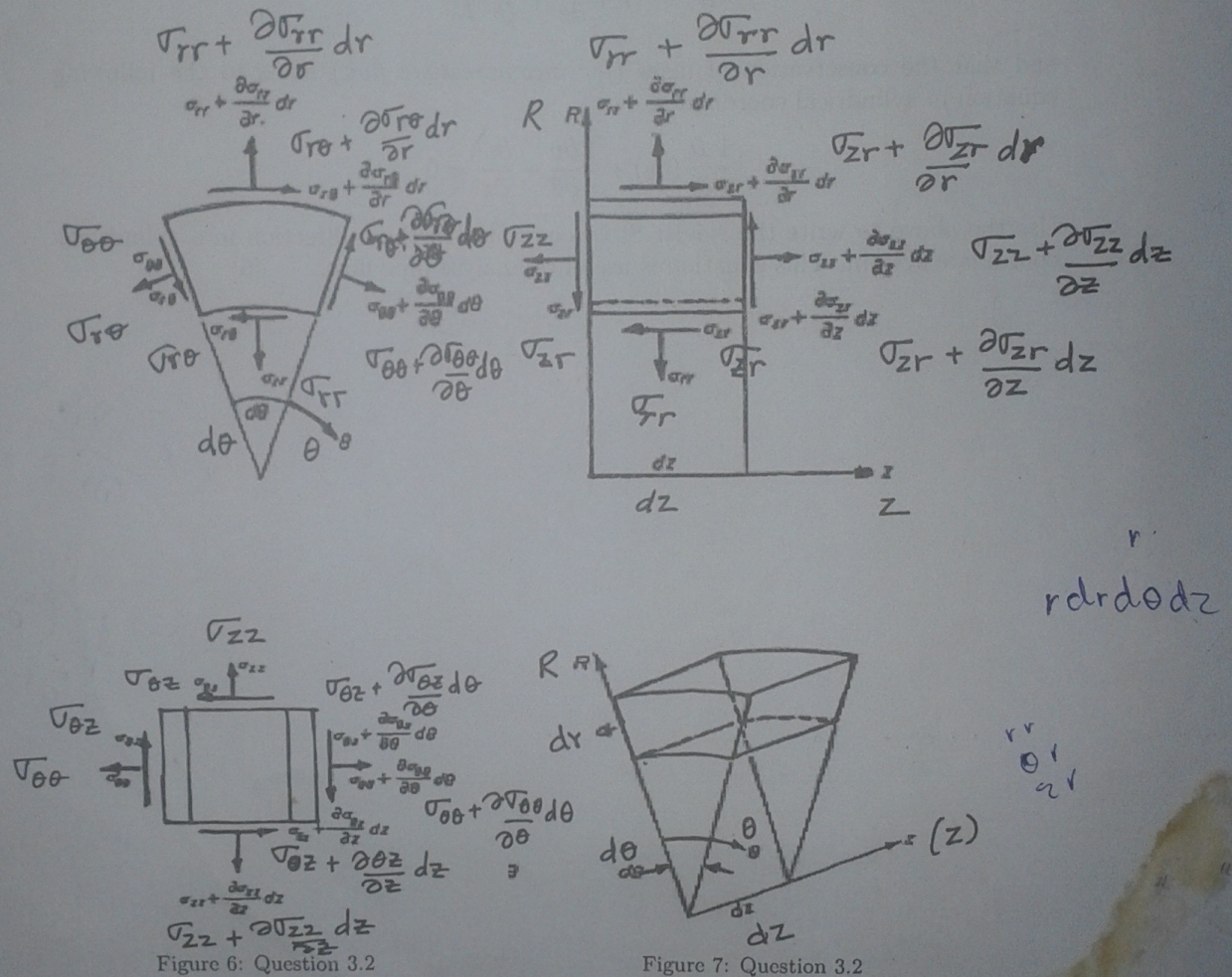
1. The first case shows fluid moving down an inclined plane. The domain is the moving fluid. This domain is open to atmosphere.

2. The second and third shows fluid moving over a stationary surface. The domain is moving fluid and this domain extends to infinity.

Write down clearly the complete, in entirety the statement of the mathematical problem to be solved to obtain the unknown quantities of interest in each of the three problems. Also, clearly state the possible difference between the governing equation in region (1) and (2) in figure (3) with the appropriate reason for this difference. [7]

Suggestion: It is possible to be very brief and succinct here. Convey the essence, use vector notation. You should save a lot of time whilst getting done with this question.

3.2 The governing differential equations in cylindrical coordinates



Our final objective in this question is to obtain the Navier Stokes equation for incompressible flow in cylindrical coordinates. We can obtain this for r , θ and z direction. However, you should work out details only for the r direction. Several information is provided to enable you to reach the equations.

Study carefully the different views of the isometric view of a piece isolated in the cylindrical coordinate system. The stresses on all the faces have been shown. Assume that this is a statics problem. Obtain first the governing partial differential equation by a force balance (in the r direction)

Further, it is given to you that the Newtonian constitutive equation in cylindrical coordinates is given as (note that u , v and w are velocities in the r , θ and z directions)

respectively.)

$$\sigma_{rr} = -p + 2\mu \frac{\partial u}{\partial r}$$

$$\sigma_{\theta\theta} = -p + 2\mu \left(\frac{u}{r} + \frac{1}{r} \frac{\partial v}{\partial \theta} \right)$$

$$\sigma_{zz} = -p + 2\mu \frac{\partial w}{\partial z}$$

$$\sigma_{r\theta} = \mu \left(r \frac{\partial(v/r)}{\partial r} + \frac{1}{r} \frac{\partial u}{\partial \theta} \right)$$

$$\sigma_{\theta z} = \mu \left(\frac{1}{r} \frac{\partial w}{\partial \theta} + \frac{\partial v}{\partial z} \right)$$

$$\sigma_{zr} = \mu \left(\frac{\partial u}{\partial z} + \frac{\partial w}{\partial r} \right)$$

and that the conservation of mass (for incompressible flow) leads to the following equation in cylindrical coordinates,

$$\frac{1}{r} \frac{\partial}{\partial r}(ru) + \frac{1}{r} \frac{\partial v}{\partial \theta} + \frac{\partial w}{\partial z} = 0$$

Use the above to write the Navier Stokes equation in the r direction in a cylindrical coordinate system. This equation is useful to analyse pipe flow. [5]

$$r \frac{\partial^2 (v/r)}{\partial r^2}$$

$$\frac{1}{r} \frac{\partial^2 v}{\partial r \partial \theta}$$